

Homework 6

Name: Diya Jain

1. $N = 571511$ and $e = 407143$

The algorithm for the attack takes the following steps.

a. $\frac{e}{N} \approx \frac{407143}{571511} = 0.7123... = 0 + \epsilon_0$. The convergent is $\frac{A}{B} = 0$.

b. $\frac{1}{\epsilon_0} \approx \frac{1}{0.7123...} \approx 1.4037... = 1 + \epsilon_1$. The convergent is $\frac{A}{B} = 0 + \frac{1}{1} = 1$.

$$C = \frac{eB - 1}{A} = \frac{407143 \cdot 1 - 1}{1} = 407142$$

$$P(x) = x^2 - (N - C + 1)x + N$$

$$= x^2 - (571511 - 407142 + 1)x + 571511$$

$$= x^2 - 164370x + 571511, \text{ which has no integer roots.}$$

c. $\frac{1}{\epsilon_1} \approx \frac{1}{0.4037} \approx 2.47709... = 2 + \epsilon_2$. The convergent is $\frac{A}{B} = 0 + \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$.

$$C = \frac{eB - 1}{A} = \frac{407143 \cdot 3 - 1}{2} = 610714$$

$$P(x) = x^2 - (N - C + 1)x + N$$

$$= x^2 - (571511 - 610714 + 1)x + N$$

$$= x^2 + 39202x + 571511,$$

d. $\frac{1}{\epsilon_2} \approx \frac{1}{0.47709} \approx 2.09604... = 2 + \epsilon_3$. The convergent is $\frac{A}{B} = 0 + \frac{1}{2 + \frac{1}{2}} = \frac{5}{7}$.

$$C = \frac{eB - 1}{A} = \frac{407143 \cdot 7 - 1}{5} = 570000$$

$$\begin{aligned}
 P(x) &= x^2 - (N - C + 1)x + N \\
 &= x^2 - (571511 - 570000 + 1)x + 571511 \\
 &= x^2 - 1512x + 571511
 \end{aligned}$$

Finding roots $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $a=1, b=-1512, c=571511$

$$x = \frac{-(-1512) \pm \sqrt{(-1512)^2 - 4(1)(571511)}}{2(1)}$$

$$= \frac{1512 \pm 10}{2}$$

$$\frac{1512+10}{2}$$

$$\frac{1512-10}{2}$$

$$= 761$$

$$= 751$$

The polynomial is $P(x) = (x-761)(x-751)$
 $N = 761 \times 751$

2. $N = 7663, x_0 = y_0 = 2, P(x) = x^2 + 3$

$$\begin{aligned}
 x_1 &= 7 \bmod 7663 = 7 & y_1 &= 52 \bmod 7663 = 52 \\
 \gcd(1x_1 - y_1, N) &= \gcd(45, 7663) = 1
 \end{aligned}$$

$$\begin{aligned}
 x_2 &= 52 \bmod 7663 = 52 & y_2 &= 7327852 \bmod 7663 = 2024 \\
 \gcd(1x_2 - y_2, N) &= \gcd(1972, 7663) = 1
 \end{aligned}$$

$$\begin{aligned}
 x_3 &= 2707 \bmod 7663 = 2707 & y_3 &= (2024^2 + 3) \bmod 7663 = 1554 \\
 \gcd(1x_3 - y_3, N) &= \gcd(1153, 7663) = 1
 \end{aligned}$$

$$x_4 = 7327852 \bmod 7663 = 2024 \quad Y_4 = (1554^2 + 3)^2 + 3 \bmod 7663 = 4029$$

$$\gcd(|x_4 - Y_4|, N) = \gcd(2005, 7663) = 1$$

$$x_5 = 4096579 \bmod 7663 = 4537 \quad Y_5 = (4029^2 + 3)^2 + 3 \bmod 7663 = 7359$$

$$\gcd(|x_5 - Y_5|, N) = \gcd(2822, 7663) = 1$$

$$x_6 = (4537^2 + 3)^2 + 3 \bmod 7663 = 1554 \quad Y_6 = (7359^2 + 3)^2 + 3 \bmod 7663 = 7471$$

$$\gcd(|x_6 - Y_6|, N) = \gcd(5917, 7663) = 97$$

First factor is 97, second factor is 79

$$N = 97 \times 79 = 7663$$